

Examples Summary

Ex 8.4:

Based on the t-test results provided earlier, there is insufficient evidence to conclude that Filter Agent 1 (Variable 1) is more effective than Filter Agent 2 (Variable 2). In fact, the data suggests the opposite trend, as the mean value for Variable 2 (8.68) is higher than that of Variable 1 (8.25).

The one-tailed p-value obtained from the test ($P(T \leq t) \text{ one-tail} = 0.157821259$) exceeds the standard significance level of 0.05. As a result, we fail to reject the null hypothesis, which posits that Filter Agent 1 is equally or less effective than Filter Agent 2.

Ex 8.6:

To determine whether the average income of males is higher than that of females, an Independent Two-Sample T-test assuming equal variances is appropriate. This test is specifically designed to compare the means of two independent groups. A one-tailed test is suitable in this context, as the hypothesis is directional—focusing on whether male income surpasses female income.

For the test results to be valid, several assumptions must be met:

- **Independence:** The two samples must be independent, meaning there is no overlap or relationship between individuals in the male and female groups.
- **Normality:** The data in each group should be approximately normally distributed.

While the histogram shows some deviation from normality, the t-test is generally robust to moderate violations of this assumption, particularly with large sample sizes ($n > 30$).

In this case, both groups contain 60 observations.

- **Equal variances:** The variances in the two groups should be similar. An F-test for equality of variances was conducted and found no significant difference, confirming the assumption of equal variances.